EMERTECH
ELECTRONICS SYSTEMS

Fluxora by Emertech
Industrial IoT
Blade Heating System

- Connect Control Optimize

One connected platform of sensors, controllers and software that monitors, controls and records every blade heating cycle, with no change to your existing PID panels.



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From manual records to controlled, traceable heating

A centralized heating system designed for precise temperature control, overheat protection, and operator-wise batch reporting.

01
LIVE MONITORING
Every panel and blanket on one dashboard.

02
SOP-LOCKED HEATING
Operators cannot alter the set profile.

03
OVERHEAT PROTECTION
Current monitored on each blanket.

04
FULL TRACEABILITY
Operator-wise batches and PDF/Excel reports.

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Multiple shop-floor problems.
One complete solution.

From manual processes and limited visibility to full control and traceability.

01
Data visible only on the panel

Today:
Temperature is shown only on the local PID and recorded on paper.

With our system:
Monitor and control every panel from any device on the network. Temperature and alarm status log automatically.

02
Alarms missed on a noisy floor

Today:
Panel alarms are easy to miss and target temperature may not be reached.

With our system:
Software alarms highlight the affected blanket, log high-temperature events and notify operators.

03
Cycle time lost after a power cut

Today:
Cycle duration is tracked by hand. Power interruption forces manual recalculation.

With our system:
Each profile step and total cycle is timed automatically, recalculated after power loss, with an alarm if a blanket is removed.

04
Set values changed at will

Today:
Operators set temperature by hand from the PID keypad, with no control over heating time.

With our system:
Predefined profiles follow your SOP. Once a profile is chosen, the PID keypad is locked and operators cannot edit it.

05
Overheating risk to the blade

Today:
Heater current is not monitored. A sensor or circuit fault can overheat a blanket and is noticed only when it is removed.

With our system:
Per-blanket energy monitoring checks load at each profile stage. Abnormal current stops heating and alerts operators.

06
No accountability or old records

Today:
No way to tell who ran a past cycle, and old data cannot be retrieved.

With our system:
Every batch is linked to its operator and archived. Reports download as Excel or PDF at any time.

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SYSTEM ARCHITECTURE

Wireless control, compatible with Multiple panels

A simple, scalable architecture that works with your multiple PID controllers.



Reach every panel along the blade

Access points linked by Ethernet (CAT 5 or higher) extend the secure Wi-Fi from panel to panel. The reference layout spaces them 100 m apart. Central monitoring needs no change to your core industrial setup.

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OUR HARDWARE

Secure, plant-ready edge hardware and server

Rugged, reliable and built for industrial environments.



Data Collector Node (DCN)

Joins legacy industrial systems to modern IoT.

- Built on the ESP32 microcontroller
- Speaks Modbus, PROFIBUS, TCP/IP, UDP and other industrial protocols
- Connects many machines on one looped RS485 port
- Encrypted wireless data and authenticated devices
- Standalone backup server and data storage if the Wi-Fi network is down
- External antenna for long-range communication in industrial areas
- Auto-registers on the configured secure network
- Multiple alarm and output configurations

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LOCAL SERVER (SBC)

The engine behind continuous operation.

Hosts the application and database, manages DCNs, and provides a secure, reliable and offline capable environment for shop-floor use.



Application & database
Hosts the system software and database.

User management
Identifies and autenticates every DCN.

System control
Manages SSL/TLS certificates, encryption keys and real-time data, graphs and reports.

Reliable operation
Local data storage (offline), automatic reconnection, and rugged hardware for harsh industrial environments.

INDUSTRIAL CONNECTIVITY

Built for secure, reliable operation.

Multiple protocols
Modbus RTU (RS485), PROFIBUS (optional) TCP/IP, UDP and more.

Encrypted communication
User autentication and access control.

Seamless integration
Works with existing PID panels.

Secure by design
SSL/TLS support and secure device registration.

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SOFTWARE & TRACEABILITY

Every cycle. Every detail.

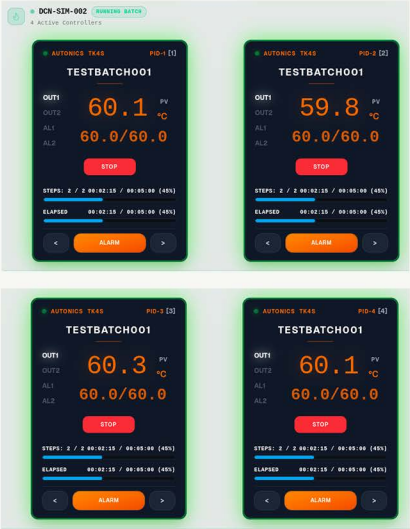
Live monitoring, process trends and batch records in one connected platform.

01 Monitor
Temperature, current and energy in real time.

02 Control
SOP-based profiles, cycle stages and alarms.

03 Trace
Operator-linked batches and historical records.

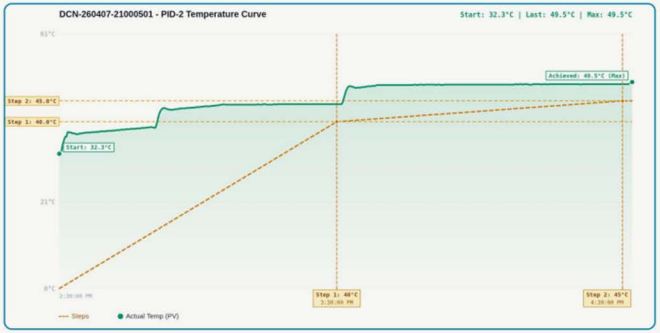
04 Export
PDF and Excel reports for quality review.



Live monitoring dashboard

- Real-time temperature, setpoint and status for every blanket
- Cycle stage and progress
- Alarm indication and notifications
- Control functions (start/stop/profile)
- Color-coded status for quick view

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Batch report (PDF/Excel)

Batch ID	24C001	**Blanket No.**	W16G001
Operator	Ramesh Kumar	**End Time**	12 Mar 2024 12:40
Start Time	12 Mar 2024 09:25	**Profile**	BLADE_HEAT_V1
		Result	Completed

Zone	Start (°C)	Peak (°C)	Soak (min)	Energy (kWh)	Status
Z1	25.1	45.2	60	12.5	OK
Z2	24.8	45.0	60	12.7	OK
Z3	25.0	45.1	60	12.6	OK
Z4	25.2	45.3	60	12.8	OK

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NEXT STEPS

Towards smarter, more reliable blade manufacturing.

Build on proven technology to improve quality, reduce risk and enable scalable production.

1 — 2 — 3

Blade heating automation

Centralized monitoring, profile control and alarms.

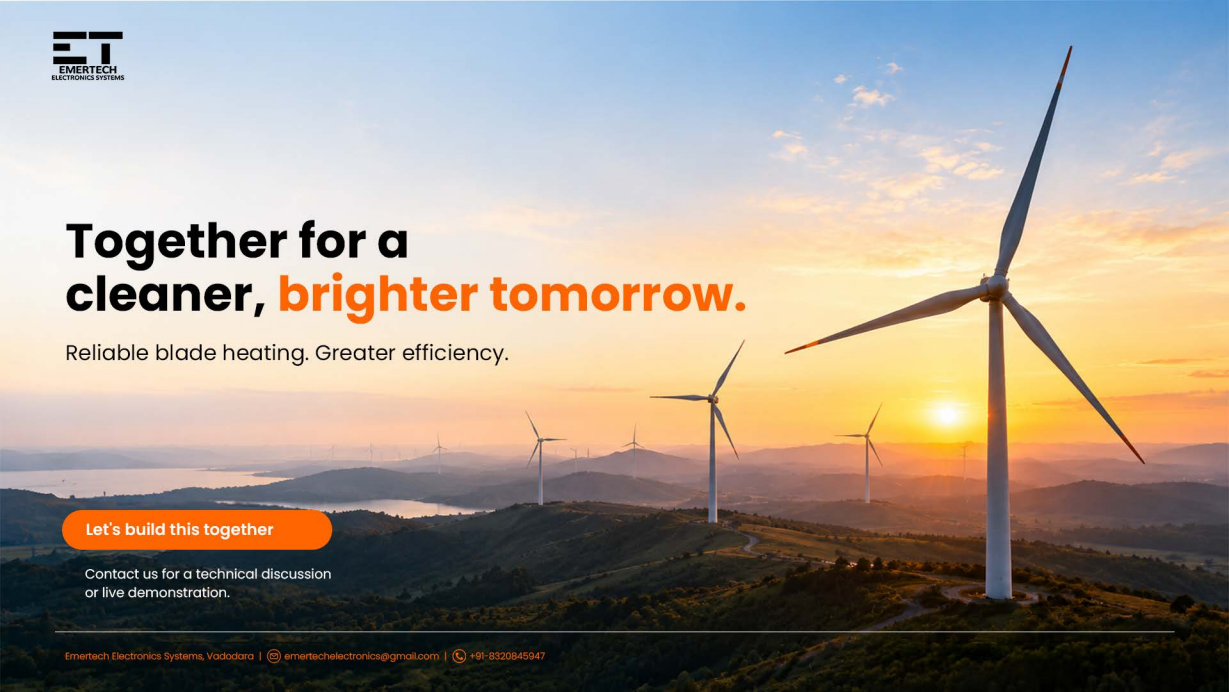
Production integration

Connect to manufacturing stations and quality checkpoints.

Scale with confidence

Higher productivity, lower risk and complete digital traceability.

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ET
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Together for a cleaner, brighter tomorrow.
Reliable blade heating. Greater efficiency.
Let's build this together
Contact us for a technical discussion or live demonstration.
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